

**A SUMMER INTERNSHIP REPORT
ON
DIGITAL MARKETING & ADVERTISING PERFORMANCE
ANALYSIS USING DATA ANALYTICS**

**Submitted To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

**In partial fulfilment of the requirement of degree of
Bachelor of Technology in Computer Science and Engineering**



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CANDIDATE'S DECLARATION

I, **CHANDAN KUMAR** do hereby declare that the internship report titled “Digital Marketing & Advertising Performance Analysis Using Data Analytics” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature

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CERTIFICATE OF EXPERIENCE

To Whomsoever It May Concern

This is to certify that **Chandan kumar** successfully completed an internship with **YuvaIntern** in the role of **Junior Data Analyst - Digital Marketing & Advertising**.

Throughout the tenure of this internship, Chandan demonstrated consistent dedication, professionalism, and a strong willingness to learn and grow. Their contributions were valued by the entire team at YuvaIntern, and they conducted themselves with integrity and commitment throughout the program.

We commend Chandan's hard work and wish them continued success in all future endeavors.

Certificate No: YI/2026/200399/439702

Date of Issue: October 11, 2026

FOR AND ON BEHALF OF YUVAINTERN




Kounal Gupta

Founder, [YuvaIntern.com](https://yuvaintern.com)

CHAPTER 4

PROJECT DESCRIPTION

4.1 Introduction

In the modern digital business landscape, data analytics forms the core of effective marketing strategies. This internship project centers around analyzing customer engagement metrics, performance trends, and conversion patterns across active digital advertising streams. By processing historical logs and customer touchpoints, the study surfaces strategic enhancements that improve overall campaign performance and lower user acquisition costs.

4.2 Organization Profile

YuvaIntern is an established experiential learning and talent ecosystem functioning in alignment with professional training institutes such as Henry Harvin. With regional headquarters across Noida (India), Delaware (United States), and Dubai (UAE), the organization bridges academic knowledge and corporate analytics implementation by hosting intensive internships in junior analytical capacities.

4.3 Problem Statement

Multi-channel advertising campaigns generate a high volume of messy behavioral datasets. The critical bottleneck for modern firms is the lack of structured preprocessing pipelines capable of unifying these logs. Without structured workflows, marketing budgets are spent on unoptimized funnels, leading to low click-through-rates (CTR) and high acquisition drops.

4.4 Project Objectives

The core technical and strategic objectives established during this project are:

1. To implement an automated programmatic cleaning pipeline for handling raw tabular campaign data.
2. To evaluate variations in user action data to discover factors driving conversion rates.
3. To build reusable analysis metrics that inform future marketing asset distribution.

4.5 Scope of the Project

The operational scope involves an analytical review of marketing records spanning a 6-week operational window. It evaluates standard digital KPIs including Click-Through-Rate (CTR), Cost Per Click (CPC), Cost Per Acquisition (CPA), and overall Return on Ad Spend (ROAS). The conclusions provide tactical optimization paths for target audiences.

4.6 Technologies and Tools Used

The technical tools deployed to execute this project include:

- Programming Environment: Python 3.x Ecosystem
- Processing Core: Pandas and NumPy Data Structures
- Visual Generation: Matplotlib and Seaborn Libraries
- Data Preparation: Advanced Microsoft Excel (Pivot tables, trend analysis charts, lookup operations)

4.7 System Architecture

The conceptual workflow of the analytical platform comprises four logical processing layers:

1. Raw Ingestion Layer: Fetching flat logs from ad managers.
2. Transformative Cleaning Layer: Processing missing data and formatting time metrics natively via Pandas arrays.
3. Compute Core: Running correlation tests between targeted demographics and performance indicators.
4. Reporting Interface: Outputting structured arrays and strategic summary decks for business execution.

4.8 Methodology

The approach follows data science lifecycle principles:

- Step 1: Ingestion of records, identifying columns containing irregular formatting.
- Step 2: Implementation of grouping models to compare ad groups and content layouts.
- Step 3: Statistical review of funnel drop-offs to pinpoint underperforming conversion steps.

4.9 Expected Outcomes

The implementation yields automated dashboards that systematically identify high-value consumer groups. Expected modifications include a 12-15% increase in funnel efficiency alongside structured cost mitigation by reducing spending on underperforming content versions.

4.10 Certificates of Completion and Proofs

Official records certifying the execution of this internship—such as the YuvaIntern Experience Certificate (ID: YI/2026/200399/439702) and Offer Letter—are attached as structured records within Chapter 3 of this document.

CHAPTER 5

BIBLIOGRAPHY / REFERENCES

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